

Technical Note

INORGANIC FILLER MODIFICATION

DESCRIPTION

The trend in manufactured paints and coatings or filled thermoplastic and thermoset resins is to increase the amount of inexpensive, inert inorganic fillers without affecting resultant resin properties. Incompatibilities of inorganic fillers with organic resins is caused by the hydrophilic nature of the filler surface, and this has limited the useful concentration of these fillers in many organic resin systems. Tyzor® organic titanates may be used to treat the surfaces of inorganic fillers to enhance their compatibility with polymeric organic material.¹

Inorganic fillers find application in epoxy resins and polyester resins, including alkyds, polyacrylates and polymethacrylates, furans, phenolics, casting resins and coating resins. Casting resins are liquid monomers or incompletely-polymerized polymers, usually containing catalysts or curing agents, capable of becoming hard after they are cast in molds. Coating resins are liquid monomers or incompletely polymerized polymers, generally in a solvent or non-solvent extender, that are capable of application by brush, roller, spray or dipping. These resins are thermosetting resins that are liquid under application conditions. Examples include paints, varnishes, enamels and lacquers.

Fillers may be particulate or fibrous and of varied shape or size, and generally reactive with the hydrolyzable group of Tyzor® organic titanates.

Examples of inorganic reinforcing materials include metals, clay, carbon black, calcium carbonate, barium sulfate, silica, mica, glass and asbestos. Reactive inorganic materials include the metal oxides of zinc, magnesium, lead, calcium and aluminum, as well as iron filings and turnings.

Examples of inorganic pigments include titanium dioxide, iron oxides, zinc chromate and ultramarine blue. Examples of organic pigments include phthalocyanine blue, quinacridone yellow, iron blue and naphthol blue.

As a practical matter, the particle size of the particulate materials should not be greater than 1 mm, preferably from 0.1 to 500 microns. The amount of the Tyzor® organic titanate used is 0.1 to 5 parts per 100 of inorganic solid. Optimum proportions are a function of the inorganic solid, the organic titanate and the effective surface area of the inorganic solid.

The reaction of the Tyzor® organic titanate takes place on the surface of the inorganic filler. The hydrolyzable group of the Tyzor® organic titanate splits off and an organic hydrophobic or oleophilic surface layer is formed on the inorganic solid. The unmodified solid, in the absence of the titanates, is difficult to disperse, especially in high concentrations, in an organic medium because of its hydrophilic surface.

The Tyzor® organic titanate may be reacted first with the inorganic and thereafter admixed with the paint or resin system. The reaction with the reactive groups on the Tyzor® organic titanates may be carried out neat or in an organic medium to form a liquid, solid or paste-like solid dispersion that can be used in compounding the final system. Such dispersions are generally stable.

Alternatively, the Tyzor® organic titanium compound may be incorporated into an organic medium (low molecular weight liquids or higher molecular weight polymeric solids) with the inorganic solid. However, the Tyzor® may react with reactive groups on the polymer and result in increased viscosity.

The following Tyzor® organic titanates are suitable depending on the nature of the system involved: Tyzor® AA-75, AA-105, TE, TOT and TNBT.

REFERENCES:

1. U.S. Pat. Nos. 3,697,474 and 3,697,475.



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